

Phase 2 Backend Development Report

Project: Learning Management System (LMS)

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Executive Summary

This report outlines the successful completion of Phase 2 backend development for the LMS platform. The primary focus was on enhancing API performance, optimizing database queries, implementing strict security measures, and preparing the system for production deployment.

Task 1: Search & Filter APIs

- Implemented advanced search functionality across Users, Courses, Enrollments, and Lessons.
- Added robust pagination and sorting features to ensure APIs remain highly scalable with large datasets.
- Optimized query structures using Prisma for dynamic filtering by category, level, and keywords.

Task 2: Approval Workflow & Analytics APIs

- Developed backend logic for robust course and instructor approval workflows.
- Created expansive analytics APIs to serve dashboard data, tracking enrollment statistics, platform-wide revenue, and course metrics.
- Overhauled media upload handlers to securely process video files and documents, increasing the upload limit to 100MB for premium lesson content.

Task 3: CORS, Rate Limiting & Error Handling

- Hardened API security by configuring strict CORS policies and Helmet HTTP headers.
- Implemented a global rate limiter to restrict excessive requests (100 requests per 15-minute window per IP), shielding the platform from DDoS and brute-force attacks.
- Engineered a structured global error handler that catches and seamlessly formats database constraints, Zod validation errors, and JWT expiration issues.

Task 4: Caching & Performance

- Integrated an in-memory caching mechanism to dramatically speed up read-heavy public endpoints (e.g., the Course Catalog).
- Implemented global response payload compression (gzip/deflate) to reduce bandwidth usage and lower latency for end-users.
- Validated Prisma query optimization to select only necessary fields, eliminating data over-fetching.

Task 5: Security Audit & Load Testing

- Conducted comprehensive dependency auditing to patch known vulnerabilities.
- Developed an automated stress-testing suite that verified the server's ability to handle over 6,600 requests per second.
- Verified the absolute effectiveness of the rate limiter under heavy synthetic load.
- Created deployment configuration templates (.env.example) to ensure a seamless production handoff.

Report proudly prepared and completed by the Backend Development Team: Shubham & Sanika.